

Conditionals

if statement: Executes a block of code if a condition is true.

else: Provides an alternative block if the condition is false.

else if: Checks multiple conditions sequentially.

switch statement: Compares a value against multiple cases and executes the matching block.

break: Stops execution in the switch after a case is matched.

default: Executes if no cases match.

// if-else

```
let age = 18;
```

```
if (age >= 18) {
```

```
    console.log("You can vote");
```

```
} else {
```

```
    console.log("You cannot vote");
```

```
}
```

// switch

```
let day = 3;
```

```
switch(day) {
```

```
    case 1:
```

```
        console.log("Monday");
```

```
        break;
```

```
    case 2:
```

```
        console.log("Tuesday");
```

```
        break;
    case 3:
        console.log("Wednesday");
        break;
    default:
        console.log("Other day");
}
```

switch uses **strict comparison (===)**.

Multiple case statements can “fall through” if break is missing.

if-else can handle **range comparisons**, whereas switch is better for discrete values.

Q: Difference between if-else and switch?

A: if-else can handle complex conditions and ranges, switch is cleaner for discrete, fixed values. switch is slightly faster for many discrete options because of optimized internal lookup.

Q: Can switch work with ranges?

A: No. switch works with exact matches; for ranges, use if-else.

Q: What happens if break is missing in a switch?

A: Execution “falls through” to the next case until a break is encountered or the switch ends.